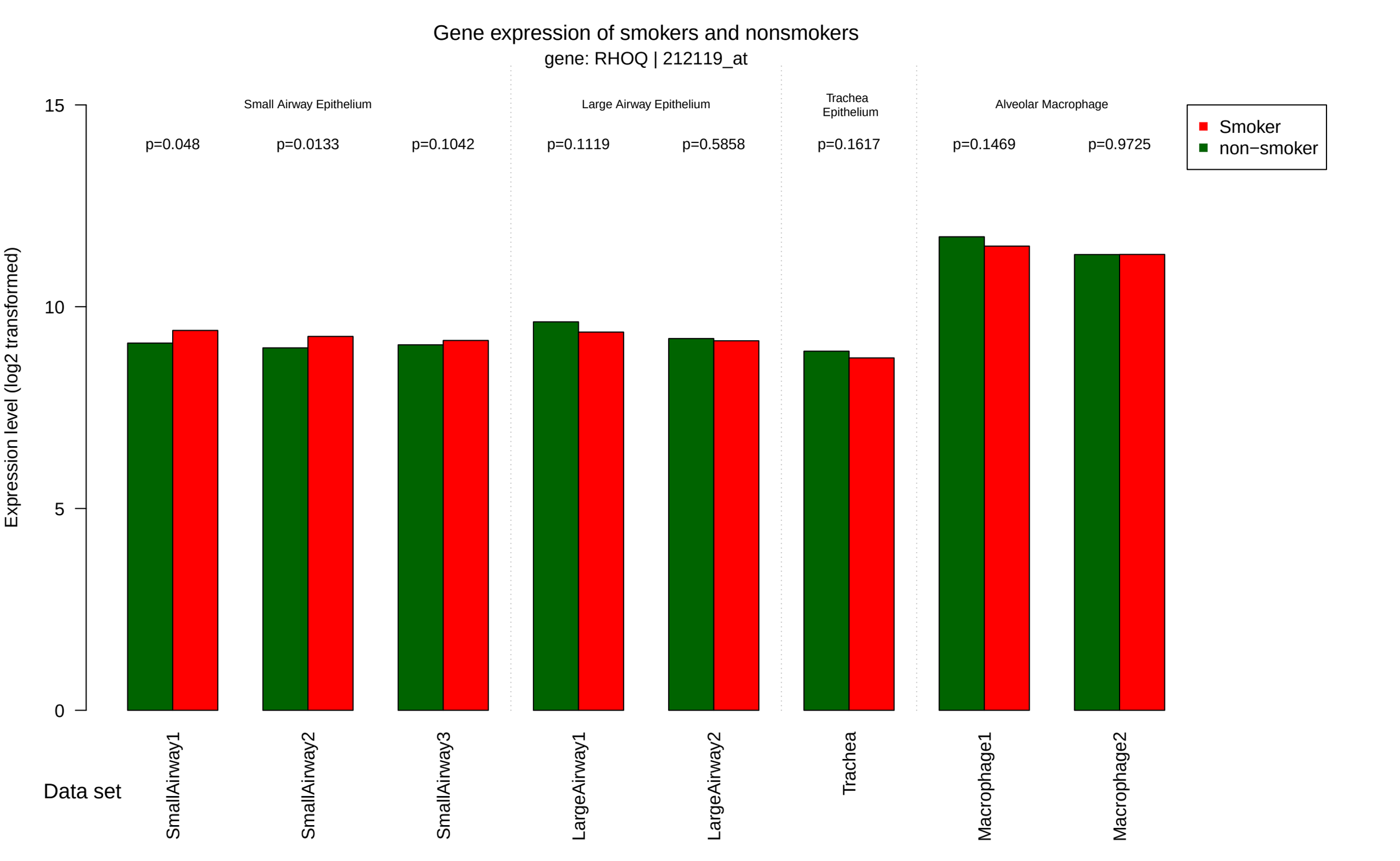
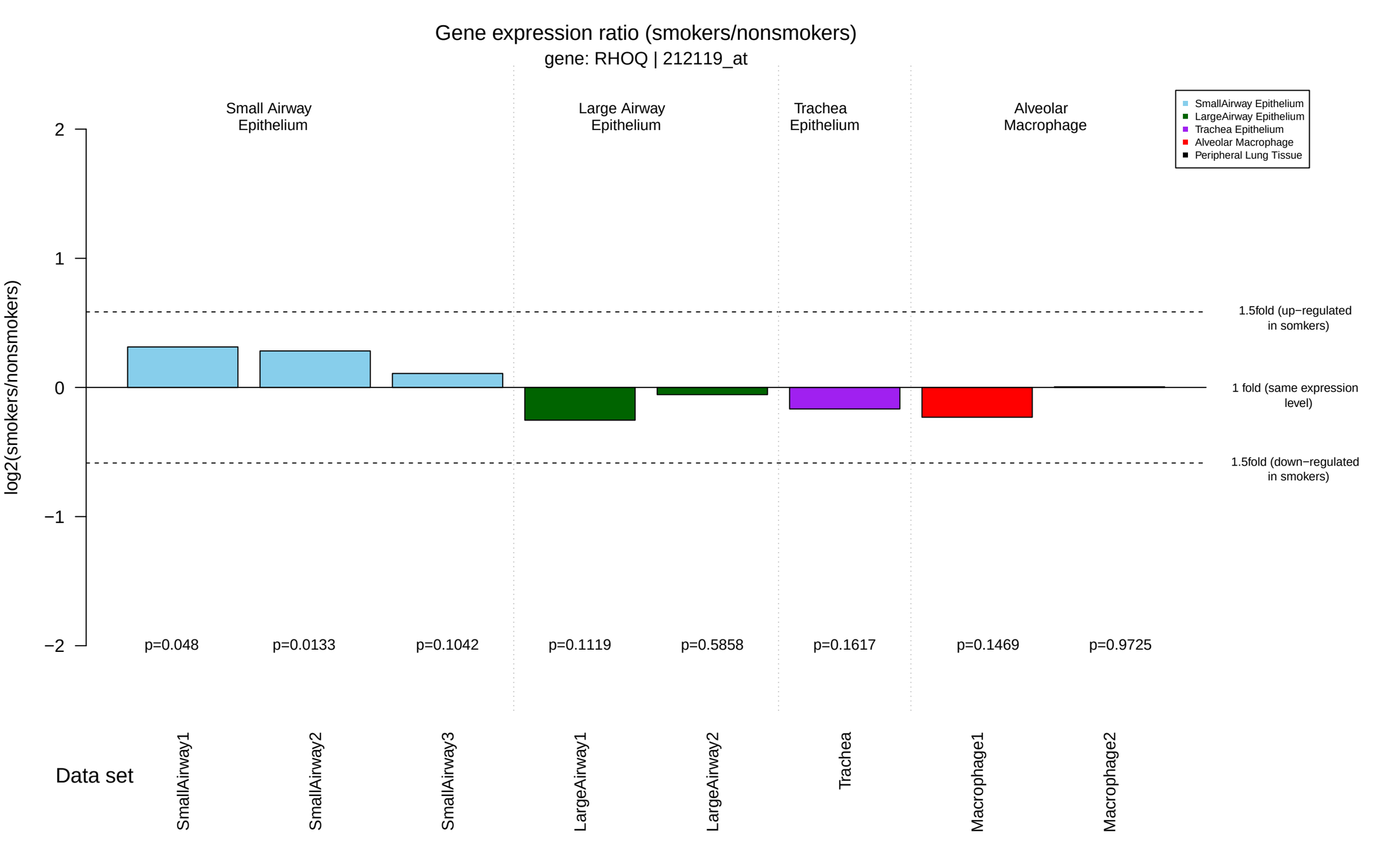
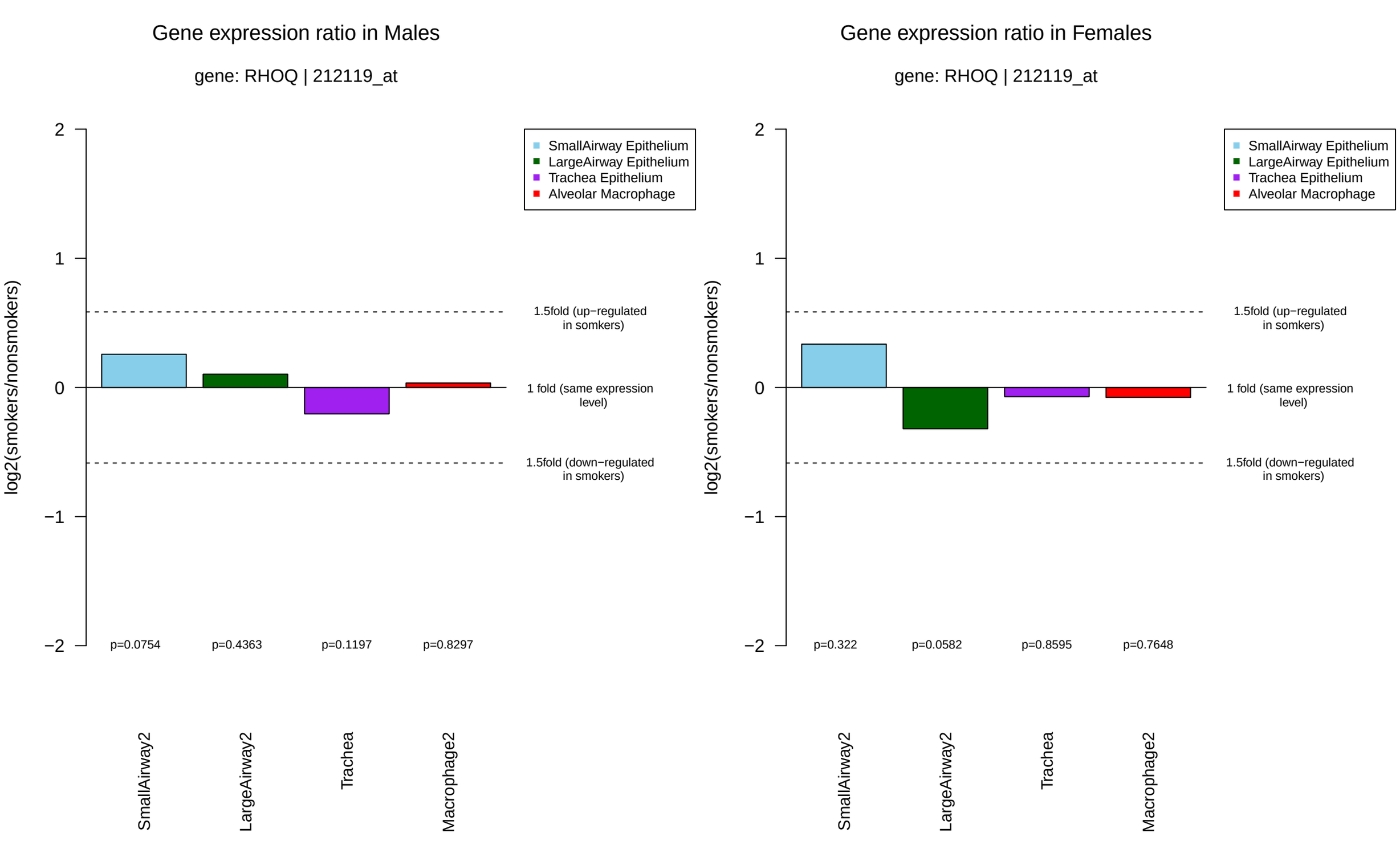
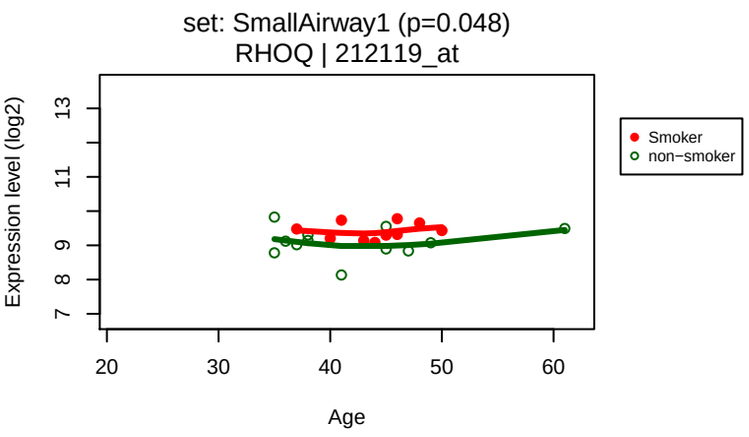
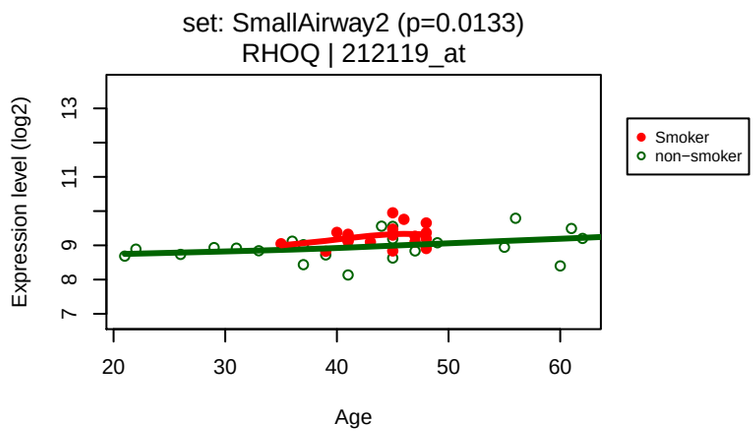
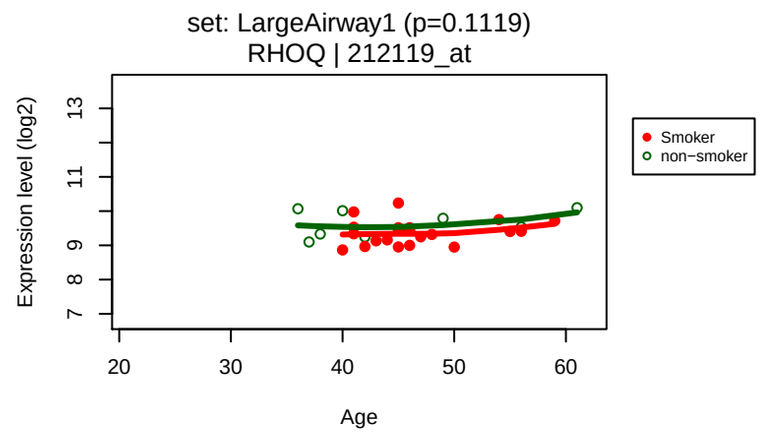

Gene expression of smokers and nonsmokers
gene: RHOQ | 212119_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.048
p=0.0133
p=0.1042
p=0.1119
p=0.5858
p=0.1617
p=0.1469
p=0.9725
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

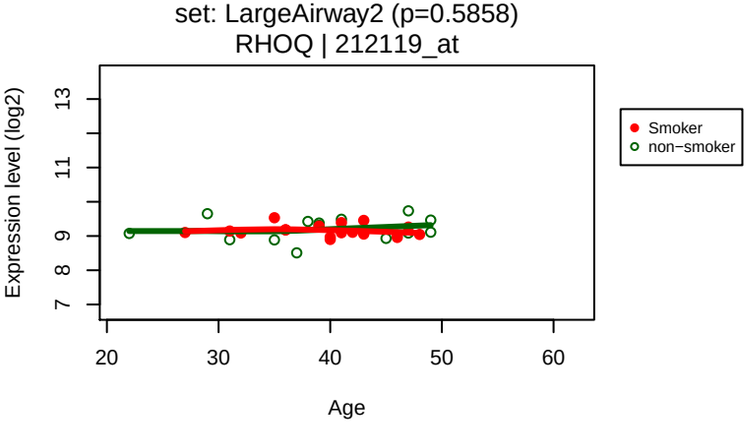

Gene expression ratio (smokers/nonsmokers)
gene: RHOQ | 212119_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.048
p=0.0133
p=0.1042
p=0.1119
p=0.5858
p=0.1617
p=0.1469
p=0.9725
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

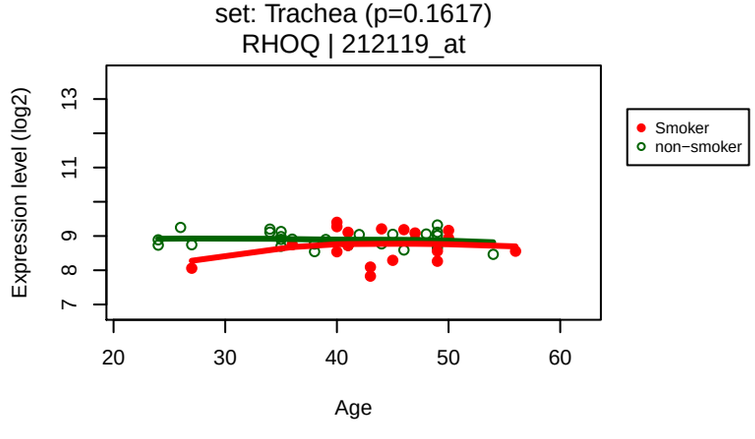

Gene expression ratio in Males
gene: RHOQ | 212119_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0754
p=0.4363
p=0.1197
p=0.8297
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOQ | 212119_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.322
p=0.0582
p=0.8595
p=0.7648
SmallAirway2
LargeAirway2
Trachea
Macrophage2

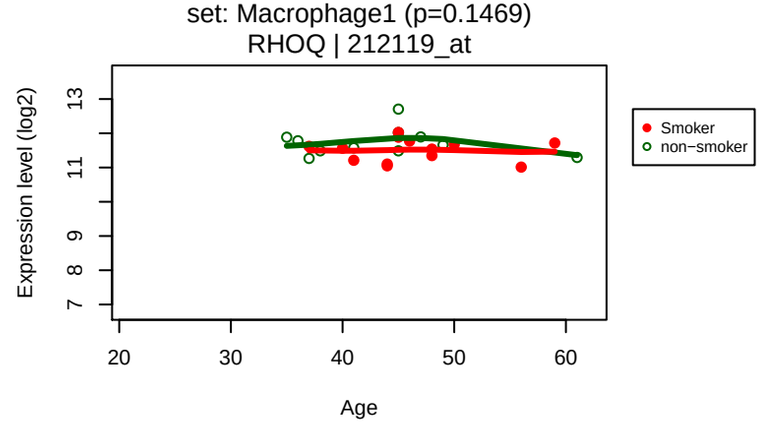

set: SmallAirway1 (p=0.048)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker

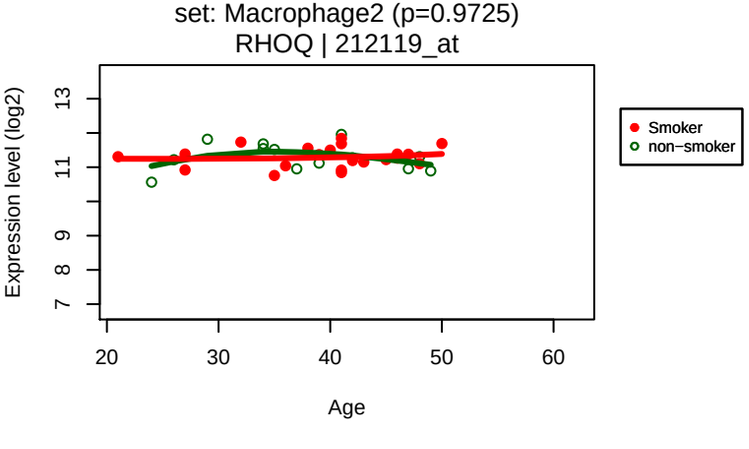

set: SmallAirway2 (p=0.0133)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker

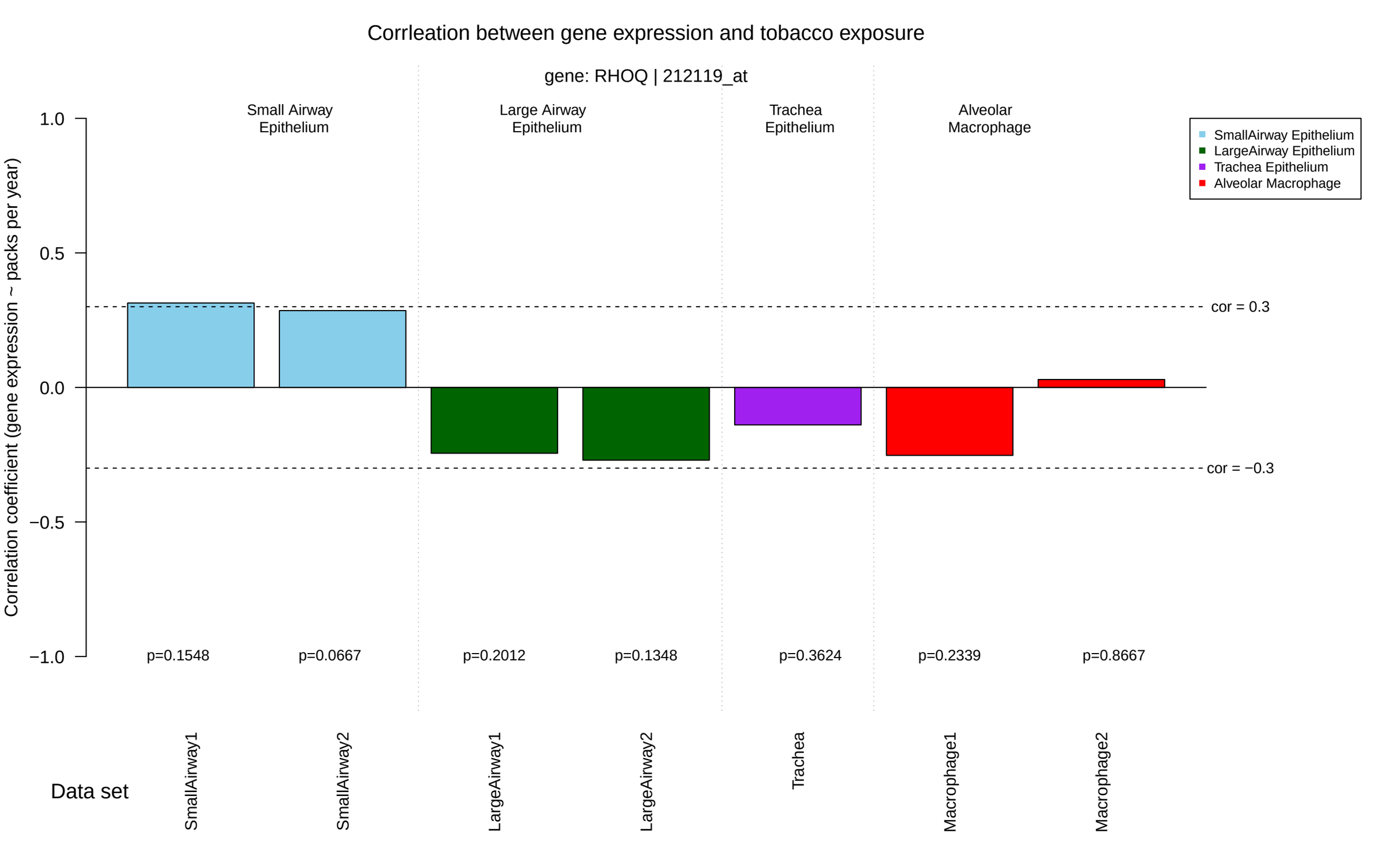

set: LargeAirway1 (p=0.1119)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5858)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1617)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1469)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9725)
RHOQ | 212119_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOQ | 212119_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1548
p=0.0667
p=0.2012
p=0.1348
p=0.3624
p=0.2339
p=0.8667
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2